

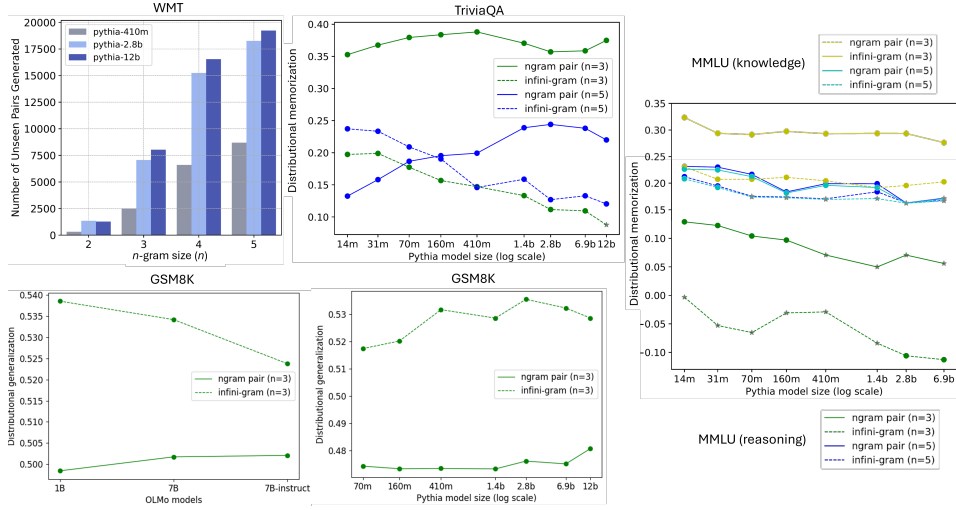


Figure 4: Visualization of distributional memorization with different-sized Pythia models on four tasks: WMT, TriviaQA, MMLU, and GSM8K. We also show results with OLMo models on GSM8K. For WMT, we show the number of new n -gram pairs generated by LLMs as the distributional memorization is not significant. For MMLU, we divide the tasks into two categories: knowledge-intensive and reasoning-intensive. For GSM8K, we show the Kendall tau ranking distance instead of Spearman correlation to quantify the distributional generalization effect as the distributional memorization is not significant. Solid lines show distributional memorization computed with our task-gram LM and dashed lines are computed with the ∞ -gram LM. Statistical significant ($p < 0.05$) values are marked with solid round markers while statistically insignificant values ($p > 0.05$) are marked with gray star markers.

in any pretraining documents in the top left panel of Figure 4. It shows that larger LLMs generate more novel n -grams, which indicates a larger discrepancy in text distribution from the pretraining data and better distributional generalization. This implies the performance increase in Figure 3 comes from a better generalization ability learned from more relevant data instead of memorization. Our hypothesis is that this generalization is the transfer of translation skills between different languages.

Such a result contradicts the observations in Merrill et al. (2024), which show that larger LLMs are less novel in n -gram generation. This is likely because they evaluate LLM generation using the prompts from an in-distribution dataset (validation set of the Pile) to the pretraining corpus, which might encourage LLMs to memorize. On the other hand, the translation dataset we use is out-of-distribution, which encourages LLMs to use its generalization capabilities.

Knowledge-intensive question-answering ability relies more on memorization. In the top middle panel of Figure 4, we show that **TriviaQA** has a significant distributional memorization effect, in terms of both task-gram LM and ∞ -gram LM. $\text{Mem}_{n=3}(\text{LLM}, \mathcal{D}|T) (> 0.35)$ is significantly more profound than $\text{Mem}_{n=5}(\text{LLM}, \mathcal{D}|T) (< 0.25)$ and $\text{Mem}_{\infty}(\text{LLM}, \mathcal{D}|T) (< 0.25)$. This indicates that LLMs memorize small, long-range parallel data pieces more than large, local data pieces.

Also, when the LM size increases, our $\text{Mem}_{n=5}(\text{LLM}, \mathcal{D}|T) (> 0.35)$ also increases. Since the larger model also has better performance as shown in Figure 3, this memorization behavior is highly correlated with LLMs’ factual QA ability. This might be because factual QA requires retrieving knowledge from training data, thus memorization plays a critical role in this task.

For **MMLU** shown in the two rightmost panels of Figure 4, we divide the 57 MMLU tasks into two groups: knowledge-intensive and reasoning-intensive. We consider knowledge-intensive tasks as tasks that can be answered by retrieving static knowledge, while reasoning-intensive tasks as ones that need computation or logical reasoning over the knowledge.⁴ In general, knowledge-intensive tasks show a more significant memorization effect than reasoning-intensive tasks. This discrepancy is the most profound when $n = 3$, while the memorization level of knowledge-intensive tasks and reasoning-intensive tasks is similar when $n = 5$. This echoes our previous hypothesis that LLMs demonstrate a stronger memorization behavior when performing knowledge-intensive tasks.

⁴The detailed classification is shown in the appendix.

Recalling rare knowledge requires generalization. Similar to TriviaQA, the top right panel of Figure 4 shows that our $\text{Mem}_{n=3}(\text{LLM}, \mathcal{D}|T)$ (> 0.25) remains most pronounced for the knowledge-intensive MMLU tasks. The primary distinction between the MMLU tasks and TriviaQA is that $\text{Mem}_n(\text{LLM}, \mathcal{D}|T)$ decreases as the LLM size increases. This may be attributed to the fact that MMLU involves more specialized and less common knowledge compared to TriviaQA, making its occurrence in the pretraining corpus relatively infrequent. Consequently, for larger models to perform better on MMLU tasks, they may need to adjust the probability of recalling this knowledge, resulting in a decrease in distributional memorization.

Reasoning-intensive abilities rely more on generalization. The reasoning-intensive MMLU tasks in the bottom right panel of Figure 4 show a very different picture compared to the knowledge-intensive MMLU tasks and TriviaQA. In this case, our $\text{Mem}_{n=5}(\text{LLM}, \mathcal{D}|T)$ is the most significant, which indicates that the memorization of large text segments is more significant. This might be because some concepts can be meaningfully expressed in large text segments while the small text segments are meaningless in a reasoning-intensive context. The decreasing trend of memorization when the model size increases also indicates that memorization is not the driving force of performance improvement.

For GSM8K shown in the two bottom panels of Figure 4, we did not observe a significant memorization effect with either Pythia models or OLMo models. To quantify the distributional generalization, we substitute the Spearman correlation with the normalized Kendall tau ranking distance, which represents the fraction of data pairs that disagree on their rankings. For both Pythia models and OLMo models, the distributional generalization increases when the model size increases, while the LLMs’ probabilities agree more with task-gram probabilities than the inf-gram probabilities. The GSM8K results confirm that generalization is the driving force of performance improvement.

Task-gram LM can better explain LLM predicted probabilities than ∞ -gram LM. With both TriviaQA and MMLU, we observe that $\text{Mem}_{\infty}(\text{LLM}, \mathcal{D}|T)$ is always less or equal to our $\text{Mem}_n(\text{LLM}, \mathcal{D}|T)$. And larger LLMs always show less ∞ -gram memorization effect. This shows that our task-gram LM is a better way to model the data distribution so that it is more correlated to what is memorized by LLMs.

In general, the LLM shows decreased memorization and increased generalization when the model size increases. This implies LLMs leverage better generalization to solve hard tasks, whether they are hard in terms of the rarity of the knowledge or in terms of the requirement of reasoning. Our task-gram LM also better models the memorized data distribution than the ∞ -gram LM.

6 INFLUENCE OF n -GRAM DATA THROUGH PRETRAINING

Since our results so far only show correlations between pretraining data frequency and the LLM predictions, we wish to explore the causal relationship of training data on LLM predictions. In this section, we estimate the influence of pretraining data on related testing predictions by accumulating the gradient dot products through model checkpoints, as introduced in Pruthi et al. (2020).

More specifically, we approximate the influence of a pretraining document at training time by the dot product between its pretraining loss gradient and the testing loss gradient. For an n -gram pair $(s^x, s^y) \in H_n(T)$ found in some example (x, y) , the test loss is defined as:

$$\ell(\theta, (x, y), s^y) = -\log P_{\text{LLM}}(s^y|u, x, y_{[1:m-1]}).$$

Here m denotes the location index of the n -gram s^y found in y , and θ denotes all trainable parameters of the LLM. The training loss of a pretraining document $d \in \mathcal{D}$ containing the n -gram pair (s^x, s^y) is defined as:

$$\ell(\theta, d, s^y) = -\log P_{\text{LLM}}(s^y|d_{[1:h-1]}).$$

Here h denotes the location index of the n -gram s^y found in d . Suppose we have k evenly spaced pretraining checkpoints of the LLM $\theta_1, \theta_2, \dots, \theta_k$, then the **influence** of d on the testing example (x, y) through the pretraining is defined as:

$$\text{In}(d, (x, y)) = \sum_{i=1}^k \sum_{(s^x, s^y) \in \Phi(x, y)} \nabla_{\theta_i} \ell(\theta_i, d, s^y) \cdot \nabla_{\theta_i} \ell(\theta_i, (x, y), s^y).$$

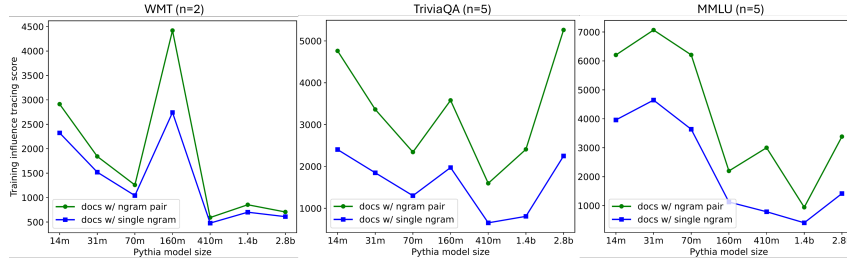


Figure 5: Training influence of pretraining documents v.s. Pythia model size with WMT, TriviaQA, and MMLU. Green lines correspond to documents containing n -gram pairs, while blue lines correspond to documents containing only the output n -gram in n -gram pairs.

	TriviaQA		GSM8K	
	Memorization	Generalization	Memorization	Generalization
Pythia (6.9B)	17%	9%	2.6%	2.8%
Pythia-Instruct (6.9B)	23.5%	23.2%	6.3%	7.3%
Pythia (12B)	28.7%	23.2%	2.7%	2.8%
OLMo (7B)	36.4%	29.8%	2.5%	3.1%
OLMo-instruct (7B)	29%	10%	6.3%	7.9%

Table 1: Zero-shot accuracy on TriviaQA and GSM8K test set with memorization encouraged task prompt (maximize counts) and generalization encouraged task prompt (minimize counts).

Here, we use $\Phi(x, y)$ to denote all n -gram pairs found in (x, y) and exist in the task-gram table $H_n(T)$. Such an influence function can be viewed as an estimation of the total reduction in loss on the test example (x, y) that is induced by the training process whenever document d is utilized in the pretraining. To estimate how much influence the documents containing n -gram pairs have on each testing task, we compute the average influence over the testing set D'_T and R randomly retrieved documents from pretraining corpus $\mathcal{D}$ for each testing example. The average influence obtained can then be written as:

$$In(\mathcal{D}, T) = \frac{1}{|D'_T|} \frac{1}{R} \sum_{(x, y) \in D'_T} \sum_{i=1}^R In(d_i, (x, y))$$

Note that the computation of the full gradient is relatively expensive, so we choose a relatively small $R = 50$. This analysis does not aim to cover the full pretraining corpus but to give complementary causal evidence to the previous findings. We consider two retrieval schemes: 1. retrieving documents where the test sample’s n -gram pair appears together. 2. retrieving documents that contain the output n -gram from the test sample’s n -gram pair. In Figure 5, we plot the average influence of these two schemes by green and blue curves, respectively.

Across all three datasets, we observe that pretraining documents containing n -gram pairs consistently contribute more to the testing examples than documents containing only the output n -gram in n -gram pairs, over different model sizes. This suggests that our task-gram table identifies important task-relevant data better than single n -grams. The difference between n -gram pair and output n -gram is the smallest on WMT, as well as the value of the influence function, which echoes the insignificant memorization effect of LLMs on this task. In general, when LLM size increases, WMT and MMLU decrease in data influence, while TriviaQA slightly increases, and has the highest influence value at the largest model size. This indicates that memorization is likely caused by more influence of the relevant data at training time.

7 PRACTICAL IMPLICATIONS: PROMPT OPTIMIZATION

An important observation of our study is that knowledge-intensive tasks benefit from LLMs’ distributional memorization, while reasoning-intensive tasks benefit from LLMs’ distributional generalization. Then it is possible to design or rewrite the prompt according to this principle to improve an LLM’s task performance, based on the hypothesis that the LLM generation distribution is strongly affected by the prompt distribution. More specifically, to encourage memorization, we can rewrite the task instruction